

5 TRIPS, 400-799 or 20600000-20620000

These cards are optional for NEW and RESTART type problems and are not used for other problem types. Two different card series are available for entering trip data, but only one series type may be used in a problem. Card numbers 401-799 allow 199 variable trips and 199 logical trips. Card numbers 20600010 - 20620000 allow 1000 variable trips and 1000 logical trips.

5.1 Trips Cancellation, 400

This card is allowed only for RESTART problems. The card causes all trips in the problem being restarted to be deleted. Any desired trips must be reentered.

W1(A) DISCARD. Any other entry is an error.

5.2 Trip Series Type, 20600000

This card, if omitted, selects card numbers 401-599 for variable trips and 601-799 for logical trips. For this case, the trip numbers are equal to the card numbers.

If this card is entered, card numbers 206NNNN0 are used for entering trip data, where NNNN is the trip number. Trip numbers (NNNN) 1-1,000 are variable trips, and 1,001-2,000 are logical trips. Trip numbers do not have to be consecutive.

W1(A) EXPANDED. Any other entry is an error.

5.3 Variable Trips, 401-599 or 20600010-20610000

Each card defines a logical statement or trip condition concerned with the quantities being advanced in time. A trip is false or not set if the trip condition is not met, and true if it is met. On restart, new trips can be introduced, old trips can be deleted, and a new trip with the same number as an old trip replaces the old trip.

The variables and parameters are the same as described for minor edits, Section 4. NULL is allowed for the right side when only a comparison to the constant is desired. The variable code TIMEOF, with the parameter set to the trip number, indicates the time at which the trip was last set. If the trip goes false, TIMEOF is set to -1.0.

W1(A) Variable. On RESTART problems, this word can also contain DISCARD or RESET. DISCARD deletes the trip; RESET sets the trip to false. If DISCARD or RESET are entered, no further words are entered on the card.

W2(I) Parameter.

W3(A)	Relationship. This may be either EQ, NE, GT, GE, LT, or LE, where the symbols have the standard FORTRAN meaning. Do not enter periods as part of the designator. For example, use GE rather than “.GE.” to specify <i>greater than</i> or <i>equal to</i> .
W4(A)	Variable.
W5(I)	Parameter.
W6(R)	Additive constant.
W7(A)	Latch indicator. If L, the trip once set true remains true, even if the condition later is not met. If N, the trip is tested each time advancement.
W8(R)	Timeof quantity (s). This word is optional. If it is not entered, the trip is initialized as false and the associated TIMEOF quantity is set to -1.0. If -1.0 is entered, the trip is initialized as false. If 0 or a positive number is entered for TIMEOF, the trip is initialized as true. TIMEOF must not be greater than 0 for NEW problems and must not be greater than the time of restart for RESTART problems.
W9(A)	Auxiliary Input for Trip Message. This word is optional. If it is not entered, no message will be displayed. It must not greater than 24 character.

The logical statement is “Does the quantity given by Words 1 and 2 have the relationship given by Word 3 with the quantity given by Words 4 and 5 plus Word 6?” If the relationship is false, the trip is false or not set. If the relationship is true, the trip is true or set. The TIMEOF variable is -1.0 if the trip is false. If the trip is true, this variable is the time the trip was last set true. A latched trip is never reset, so the trip time never changes once it changes from -1.0. For the nonlatched trips, the trip time when set remains constant until the trip condition becomes false and then the trip time is -1.0 again. If the trip condition becomes true again, the process is repeated. For trips such as a time test, L should be used to eliminate repeated testing, although no error or difference in results will occur if N is used.

5.4 Logical Trips, 601-799 or 20610010-20620000

If these cards are entered, at least one of the variable trips must have been entered. Each card defines a logical relationship with the trips defined on these cards or on the variable trips.

W1(I)	Trip number. The absolute value of this number must be one of the trip numbers defined by the variable or logical trips. A negative trip number indicates that the complement of the trip is to be used in the test.
W2(A)	Operator. The operator may be AND, OR, or XOR. For RESTART problems, this quantity may also contain DISCARD or RESET. DISCARD deletes the trip and RESET

sets the trip to false. If DISCARD or RESET are entered, no further words are entered on the card and Word 1 (W1) may be 0.

- W3(I) Trip number. This is similar to Word 1 (W1).
- W4(A) Latch indicator. If L, the trip when set remains set. If N, the trip is tested each time advancement.
- W5(R) Timeof quantity (s). This word is optional. If not entered, the trip is initialized as false, and the associated TIMEOF quantity is set to -1.0. If -1.0 is entered, the trip is initialized as false. If 0 or a positive number is entered for TIMEOF, the trip is initialized as true. TIMEOF must not be greater than 0 for NEW problems and must not be greater than the time of restart for RESTART problems.
- W6(A) Auxiliary Input for Trip Message. This word is optional. If it is not entered, no message will be displayed. It must not greater than 24 character.

The trip condition is given by the result of the following logical expression:

CONDITION OF TRIP IN W1 OPERATOR CONDITION OF TRIP IN W3.

5.5 Trip Stop Advancement, 600

This card can be entered in NEW and RESTART problems. One or two trip numbers may be entered. If either of the indicated trips are true, the problem advancement is terminated. These trips are tested only at the end of a requested advancement. If the trips can cycle true and false, they should be latched-type trips to ensure being true at the test time.

- W1(I) Trip number.
- W2(I) Trip number. A second trip number need not be entered.